

Emma Fries, EIT

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Education

California Polytechnic State University - San Luis Obispo, CA

June 2024

Bachelor of Science in Mechanical Engineering

Engineer-in-Training (#186647) - California Board for Professional Engineers

December 2025

SolidWorks Certifications: CSWA (October 2025), CSWP (In Progress), CSWA-S (In Progress)

Skills

Design & Analysis: SolidWorks, FEA (SolidWorks Simulation), GD&T, MATLAB, Python, AutoCAD, Microsoft Excel

Fabrication: 3D Printing (FDM), Prototyping, Manual Machining (Mill, Lathe), CNC Machining (Introductory)

Engineering Project Work

Visual Inspection System Project - Edwards Lifesciences (NDA)

September 2023 – June 2024

- Collaborated on a student team to design and prototype an automated vision system to detect manufacturing defects in heart valve components, delivering a proof-of-concept prototype demonstrating the feasibility of an automated inspection system
- Designed custom pneumatic fixtures, Meca500 robot motion control, and optical inspection assembly, ensuring precise integration of mechanical and robotic systems
- Modeled and designed components in SolidWorks and applied GD&T standards to technical drawings for fabrication via 3D printing and manual machining, supporting efficient manufacturing and assembly
- Implemented robot motion and pneumatic control using RoboDK and Mecademic software, integrating Basler Pylon image acquisition to generate high-precision inspection images
- Developed system verification documentation, including Preliminary, Critical, and Final Design Reviews, testing procedures with pass/fail criteria, DVP&R, and FMEA

Cross-Flow Water Turbine - California Polytechnic State University

April 2024 – June 2024

- Built and tested a Banki-Michell water turbine system that generated sufficient power to pull a 1 N load 20 feet
- Designed the turbine, nozzle, and pulley system in SolidWorks, applying maximum power transfer principles and EES calculations while working within constraints on reservoir height, water volume, and component dimensions
- Fabricated the pulley and structural test stand using woodworking tools and prototyped key components (nozzle, turbine, reservoir) via 3D printing, enabling rapid iteration
- Conducted testing and implemented design improvements, resulting in increased turbine efficiency and speed

Relevant Project Work Experience

Project Lead

March 2025 – February 2026

Lucca Delicatessen

San Francisco, CA

- Assisted management with the transition to a new inventory and point-of-sale system by defining new processes, training coworkers on system organization and the new operating model, and structuring and inputting data
- Onboarded and trained new team members, ensuring high quality service, adherence to safety protocols, and effective collaboration in a fast paced environment

Project Administrator

August 2024 – February 2025

Rubicon Partners Inc.

Sacramento, CA

- Assisted in due diligence and acquisition support for a large-scale high-rise commercial property, managing documentation, overseeing leases and contracts, and coordinating with tenants
- Developed expertise in the building's audio-visual system, becoming the go-to resource for tenants and leading tutorials to improve user proficiency and system efficiency
- Collaborated with the project team to support financial and lease operations, including invoice processing, expense tracking, and maintaining accurate documentation